

Problem

Obtain the overall impedance for the following cases:

(a) $P = 1000 \text{ W}$, $\text{pf} = 0.8$ (leading),
 $V_{\text{rms}} = 220 \text{ V}$

(b) $P = 1500 \text{ W}$, $Q = 2000 \text{ VAR}$ (inductive),
 $I_{\text{rms}} = 12 \text{ A}$

(c) $S = 4500 \angle 60^\circ \text{ VA}$, $V = 120 \angle 45^\circ \text{ V}$

Step-by-step solution

Step 1 of 3

(a)

$$S = P - jQ$$

$$S = 1000 - j \left(\frac{1000}{0.8} \right) \sin(\cos^{-1}(0.8))$$

$$S = 1000 - j750$$

But,

$$S = \frac{|V_{\text{rms}}|^2}{Z^*}$$

$$Z^* = \frac{|V_{\text{rms}}|^2}{S}$$

$$Z^* = \frac{(220)^2}{1000 - j750}$$

$$Z^* = 30.98 + j23.23$$

$$Z = 30.98 - j23.23 \ \Omega$$

Step 2 of 3

(b)

$$P = 1500 \text{ W},$$

$$Q = 2000 \text{ VAR}$$

$$S = P + jQ$$

$$S = 1500 + j2000$$

$$S = |I_{rms}|^2 \times Z$$

$$Z = \frac{S}{|I_{rms}|^2}$$

$$Z = \frac{1500 + j2000}{(12)^2}$$

$$Z = 10.42 + j13.89 \text{ } \Omega$$

[Comments \(1\)](#)



Anonymous

why is $S = P + jQ$ not $S = P - jQ$ like part A?

Step 3 of 3

(c)

$$Z^* = \frac{|V_{rms}|^2}{S}$$

$$Z^* = \frac{|V|^2}{2S}$$

$$Z^* = \frac{(120)^2}{(2)(4500 \angle 60^\circ)}$$

$$Z^* = 1.6 \angle -60^\circ$$

$$Z = 1.6 \angle 60^\circ$$

$$\boxed{Z = 0.8 + j1.386 \text{ } \Omega}$$